

Crop Recommendation Model for Food Security

Machine Learning Project Report

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Dataset: Crop Recommendation Dataset
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1. Executive Summary

This project develops a supervised machine learning model that recommends the most suitable crop based on soil nutrients and environmental conditions. The goal is to support better crop selection decisions, reduce the risk of crop failure, and connect data science to the larger theme of improving food security and sustainable agriculture.

The final model comparison found that Random Forest performed best on the test set, reaching 99.55% accuracy and a weighted F1-score of 99.55%. The strong performance is supported by the balanced dataset structure and clear relationships between crop labels and the seven input features.

2. Study Theme and Objective

Study theme: Use data science and machine learning to improve food security and reduce hunger by making agriculture more efficient, fair, and sustainable.

Project objective: Build a classification model that predicts the best crop to grow from nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. This objective directly supports the theme because choosing the right crop can improve yield potential, resource efficiency, and planning under changing climate and soil conditions.

3. Dataset Description

The project uses the Crop Recommendation Dataset, which contains 2,200 records, 7 numeric input features, and 22 balanced crop classes with 100 records per class. The target variable is the crop label.

Category	Details	Project Role	Data Type
Soil nutrients	N, P, K	Predictors	Integer
Climate/weather	Temperature, humidity, rainfall	Predictors	Float
Soil chemistry	pH level	Predictor	Float
Crop recommendation	22 crop labels	Target	Categorical

Dataset quality checks found no missing values. Each crop class has exactly 100 records, so the classification task is balanced and does not require resampling.

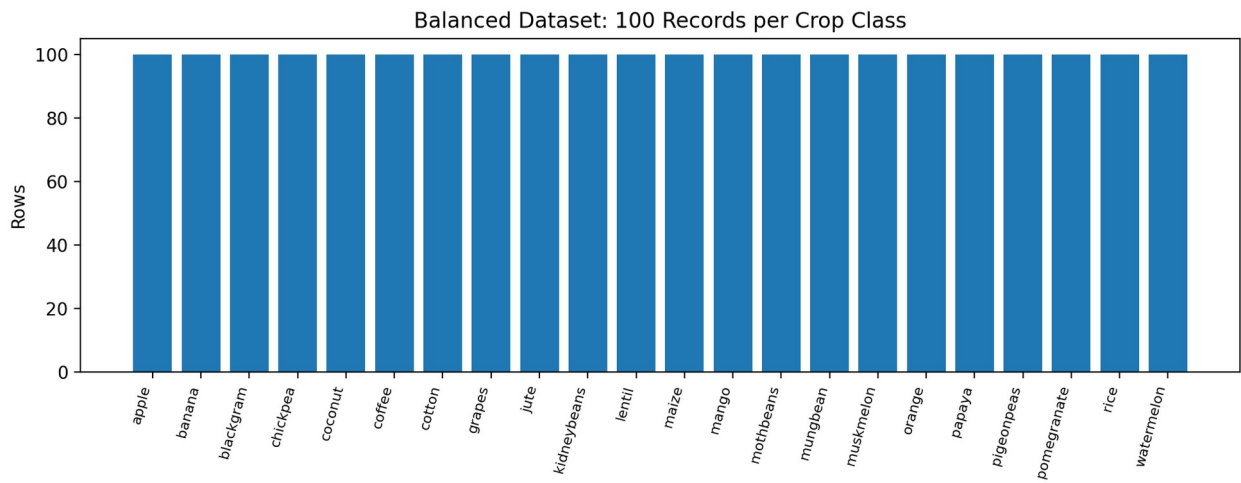


Figure 1. Crop class distribution.

4. Machine Learning Process

4.1 Data Exploration

Exploratory analysis reviewed feature ranges, class balance, and correlations between numeric variables. The dataset includes soil nutrient values, weather conditions, pH, and rainfall, which are all reasonable predictors for crop suitability.

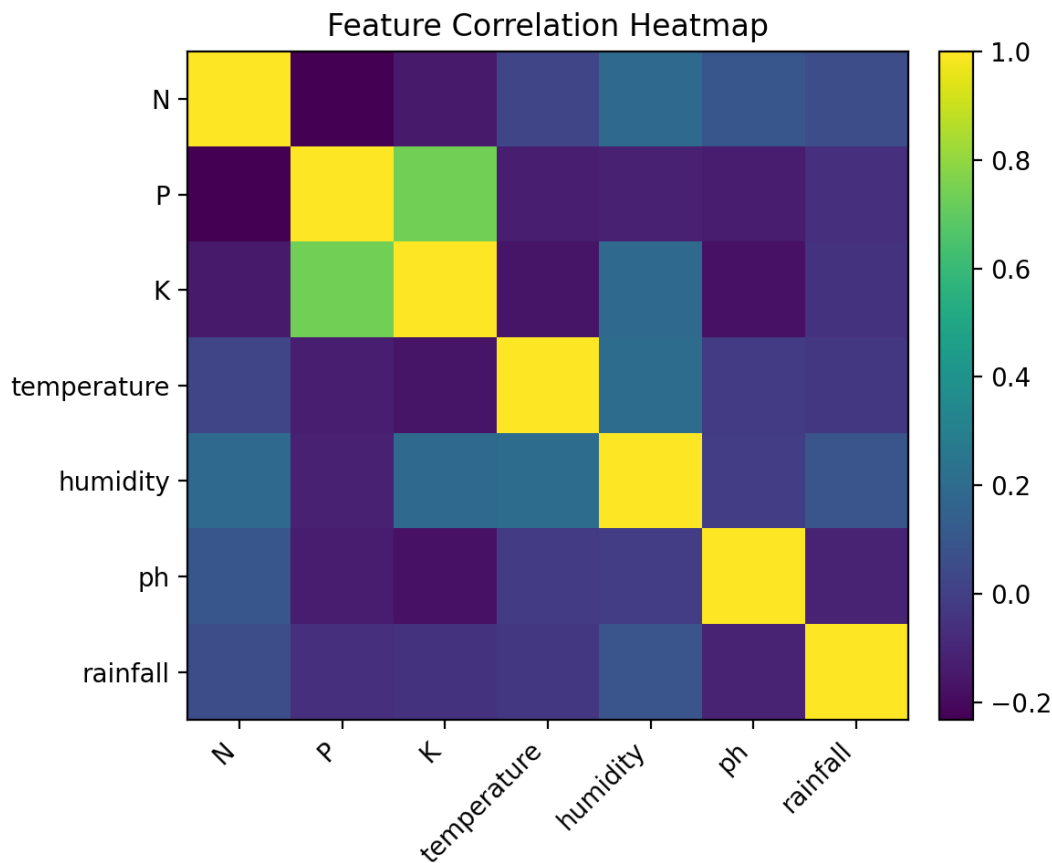


Figure 2. Correlation heatmap for numeric features.

4.2 Data Cleaning

The dataset was clean and required minimal preprocessing. Steps included checking missing values, confirming numeric data types for input features, verifying the target label, and ensuring that every class had enough samples for train/test splitting.

4.3 Feature Engineering and Selection

All seven original features were retained because each has a clear agronomic connection to crop growth. No features were removed. Standardization was applied only for distance- or margin-based models such as KNN and SVM. Tree-based models did not require scaling.

4.4 Model Selection

Four supervised classification algorithms were compared: Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector Machine. These models were selected because they represent interpretable tree logic, ensemble learning, distance-based classification, and maximum-margin classification.

4.5 Data Splitting and Validation

The dataset was split into 80% training data and 20% testing data using stratified sampling. Stratification preserved the equal crop-class distribution in both training and testing sets. Performance was measured on the unseen test set.

4.6 Performance Metrics

The models were evaluated using accuracy, weighted precision, weighted recall, weighted F1-score, and confusion matrix analysis. Weighted metrics were used because this is a multi-class classification problem.

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.9955	0.9957	0.9955	0.9955
SVM	0.9886	0.9896	0.9886	0.9887
Decision Tree	0.9795	0.9806	0.9795	0.9794
KNN	0.9795	0.9804	0.9795	0.9793

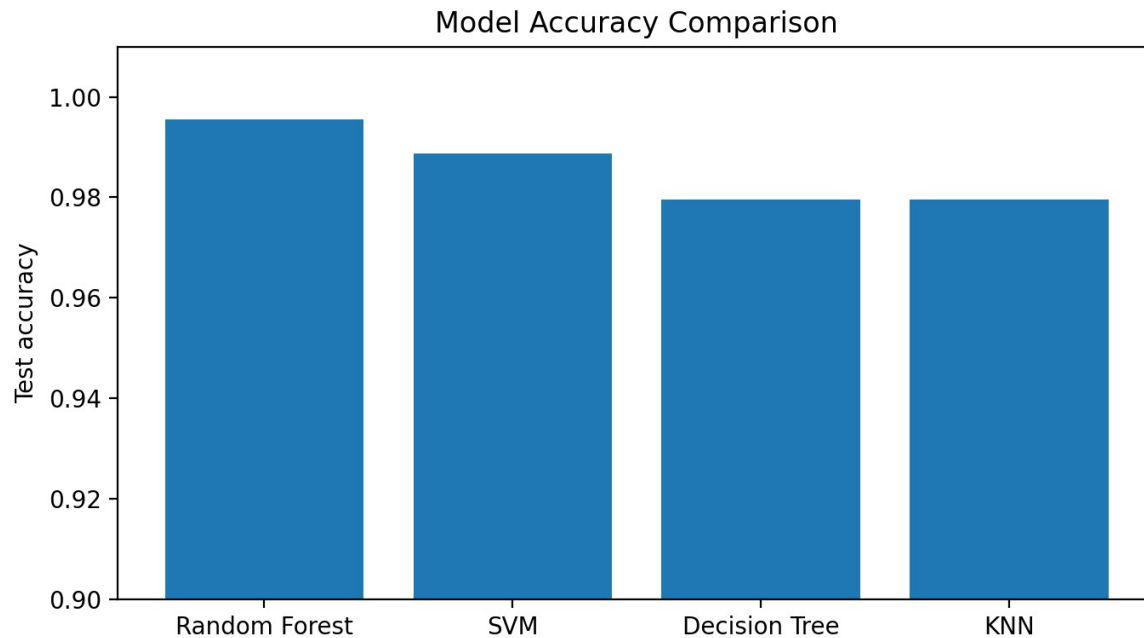


Figure 3. Model accuracy comparison.

5. Final Model Results

Random Forest was selected as the final model because it had the best overall accuracy and F1-score while also providing feature importance values that help explain the prediction process. The model performed well across the 22 crop classes, with only a small number of test-set errors.

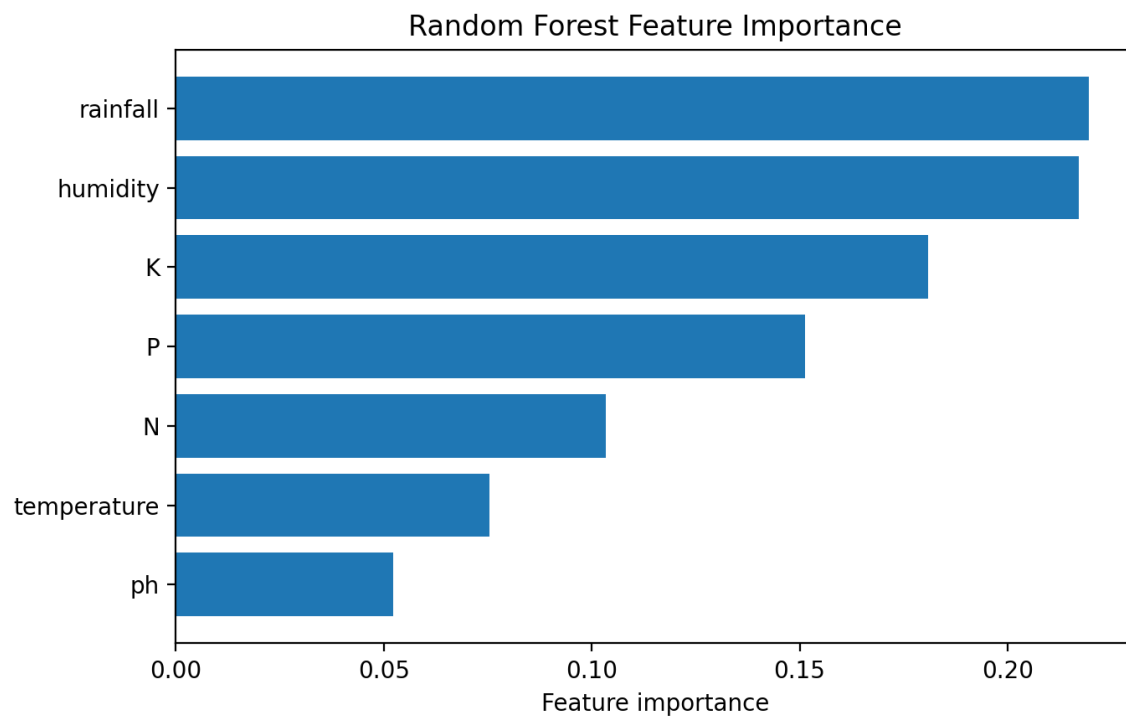


Figure 4. Random Forest feature importance.

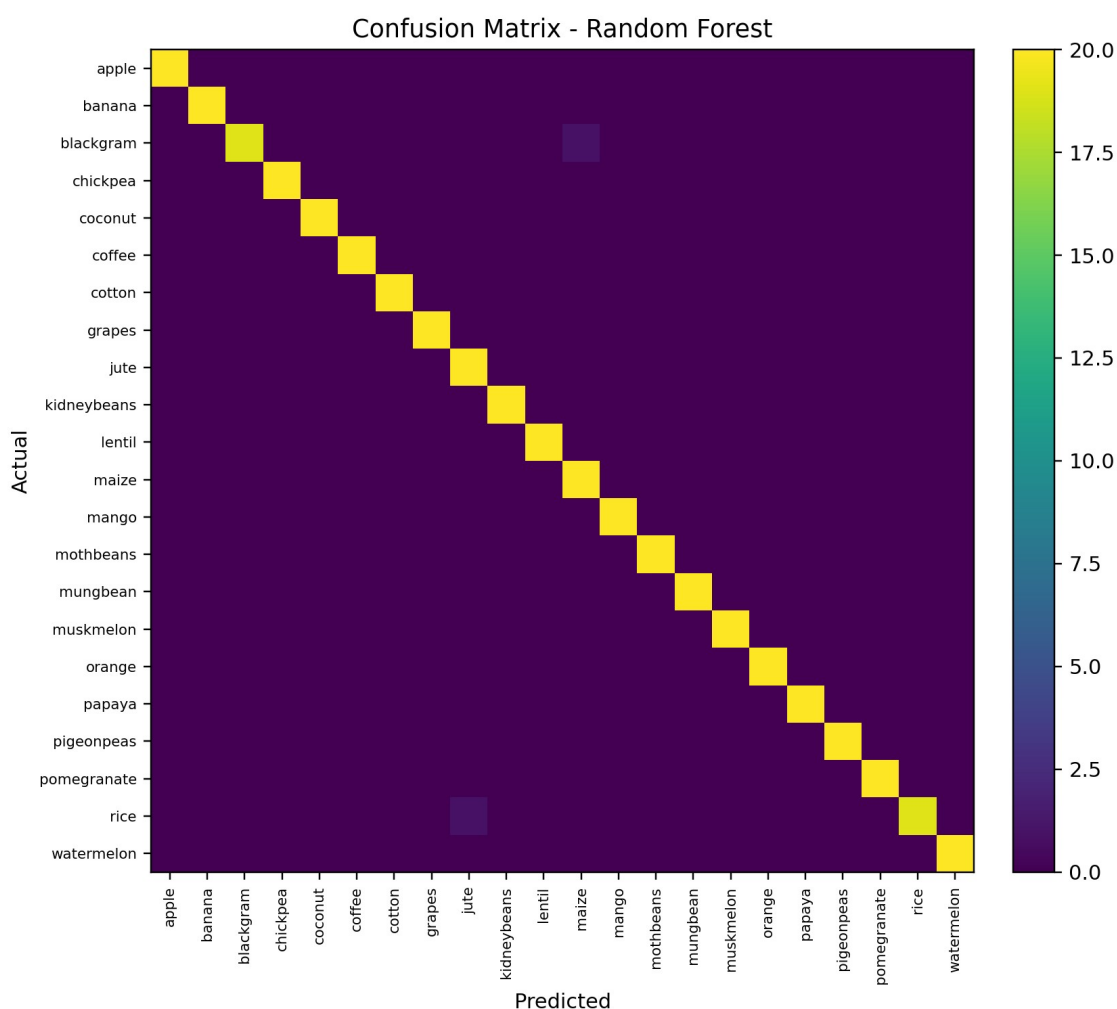


Figure 5. Confusion matrix for the best-performing model.

6. Interpretation and Practical Value

The model can be used as a decision-support tool for crop planning. Farmers or agricultural advisors can enter soil nutrient values and environmental conditions to receive a recommended crop. In real use, this type of model should be combined with local farming knowledge, market demand, seed availability, irrigation access, and risk factors such as pests or extreme weather.

The project supports food security because better crop selection can reduce waste, improve productivity, and help communities make more informed agricultural decisions. It also supports sustainability by encouraging crops that are better matched to local conditions instead of forcing unsuitable crops through excessive fertilizer, water, or chemical inputs.

7. Limitations

- The dataset is clean and balanced, while real agricultural data may be noisy, incomplete, and region-specific.

- The model does not include economic factors such as crop price, labor cost, storage, or transportation.
- Weather conditions are treated as static inputs, but real farms experience changing conditions across the growing season.
- The model should be validated with local field data before being used for real farming decisions.

8. Conclusion

This project successfully builds and evaluates a crop recommendation model using soil and environmental variables. Random Forest achieved the strongest test-set performance and was selected as the final model. The project shows how machine learning can support smarter agricultural planning and contribute to the larger goal of improving food security and reducing hunger.

9. References

1. Crop Recommendation Dataset from Kaggle, originally referenced in the project guidelines.
2. vishant007 GitHub mirror of Crop_recommendation.csv, used to access the dataset file for this completed project.
3. Course guideline documents: Module 7 project theme/objective/dataset and Module 9 machine learning process plan.